

Particulate Matter Monitoring & Health Effects

Issue 2026-08-27 | Entry window 27 Aug | 36 in scope, 53 out

36IN-WINDOW RECORDS
(53 SCREENED OUT)**10**SUBTOPIC CLUSTERS
ALL OCCUPIED**13**INTERVAL-BEARING
ESTIMATES**14**TIER-A
RECORDS**4**ABSTRACT-FREE
DEPOSITS (11%)

Signal of the day

A meta-analysis of 19 studies shows that the *harmonisation step* — not the exposure — carries most of the heterogeneity in the PM–venous-thromboembolism literature.

Pooling 19 observational studies and more than **4.5 million participants**, the headline long-term estimates are null and enormously heterogeneous: **PM_{2.5} HR 1.42 (95% CI 0.88–2.28)**, $I^2 = 97.0\%$ and **PM₁₀ HR 1.18 (0.85–1.64)**, $I^2 = 96.2\%$. Restricting to the subset of studies that *natively reported* effects per $10\ \mu\text{g}/\text{m}^3$ — i.e. dropping every estimate the reviewers had to rescale — collapses the heterogeneity to $I^2 = 0\%$ and yields **PM_{2.5} HR 1.070 (1.010–1.135)** and **PM₁₀ HR 1.023 (1.000–1.047)**. Two orders of magnitude of I^2 removed by a data-handling decision is not a subgroup finding; it says the rescaling of non-native contrasts (per-IQR, per-SD, per-quartile) into a common per-10 unit is importing variance that the underlying studies do not contain. The direct consequence for distributed sensing: a network that reports its own exposure contrast in a non-standard unit is not merely inconvenient for meta-analysis, it is *actively destabilising* to it. Reporting per- $10\ \mu\text{g}/\text{m}^3$ natively is the cheapest methodological contribution a monitoring study can make.

What changed today

- **The largest single-day issue this watch has built, and only the fourth to occupy all ten clusters.** 36 records against a median of roughly 14 — mechanical, not scientific: a *Toxics* volume release supplied six, and Crossref's by-ISSN leg 43 deposits of which most were rejected. Only the 26 July founding backfill (164 records over a month) is larger. **Four deposits arrived abstract-free**, three of them on the sensing axis — the refractive-index, fire/extreme-PM, TRAKER and Yangtze records.
- **Low-cost PM sensors used as a biological proxy, with a validated transfer function.** Li et al. treat greenhouse PM sensors not as air-quality instruments but as continuous physical proxies for airborne *Pseudomonas* aerosol load, triggering targeted qPCR only when an estimated risk threshold is crossed. Across **732 plot-days** the surveillance model reached $R^2 = 0.884$ **for pathogen aerosol accumulation** and the downstream forecast $R^2 = 0.874$ **for daily disease-index change rate**. The framing is the interesting part: a sensor with no biological specificity is made useful by pairing it with a specific but expensive assay that it schedules.
- **Condensable PM gets a real-time instrument and an age-resolved dose.** Kang et al. characterised CPM from eight vehicle types with a purpose-built real-time system (PNU-CPM). CPM accounted for **63.2% of total PM mass for an untreated level-4 diesel** against **8.2% for GDI**; organics were dominated by alkanes ($\sim 53.5\%$) and the water-soluble fraction by SO_4^{2-} and Na^+ . MPPD simulation gives higher mass-per-body-weight lung deposition in younger age groups. CPM is invisible to hot-filter source testing by construction, and semi-volatile material of this composition will not obey the scattering and hygroscopic assumptions baked into optical-sensor corrections.
- **Speciation beats mass again, on oxidative potential.** In Xi'an, XGBoost-SHAP attribution over DTT assays makes **potassium and manganese** — a biomass-burning tracer and a redox-active transition metal — the dominant contributors to volume-normalised OP, despite water-soluble metals being a minor mass fraction. Dust dominates winter OP by loading, traffic dominates summer OP by intrinsic toxicity. Same conclusion as the 26 August component-IRR result, different measurement chain.
- **A well-powered null more informative than most positives.** Instrumenting municipal PM_{2.5} with wind direction across **>3.5 million French municipality-weeks**, Morvan et al. find *no* effect on COVID-19 hospitalisation or death among people without pre-existing conditions, and a clear one once comorbidity is conditioned on (**+1.5 deaths per 100,000 per $10\ \mu\text{g}/\text{m}^3$** , 0.8–2.2; respiratory 0.9 (0.4–1.4), cardiovascular 0.6 (0.1–1.1)). The 2020–21 literature was overwhelmingly ecological and positive; this design has an instrument, and it puts the effect entirely in the susceptible stratum.
- **PM₁ outperforms PM_{2.5} in two Guangzhou cohorts.** Prenatal PM₁ gave the largest per- $5\ \mu\text{g}/\text{m}^3$ odds of a rapid fetal-weight trajectory (**OR 1.56, 1.03–2.37**, versus 1.44 for PM_{2.5} and 1.28 for PM₁₀), and the toddler-sleep cohort put the sub-micron fraction on par with PM_{2.5}. One city, one exposure vendor, so independence is partial — but the ordering is what deposition predicts, and what mass-only monitoring cannot see.

Corpus at a glance

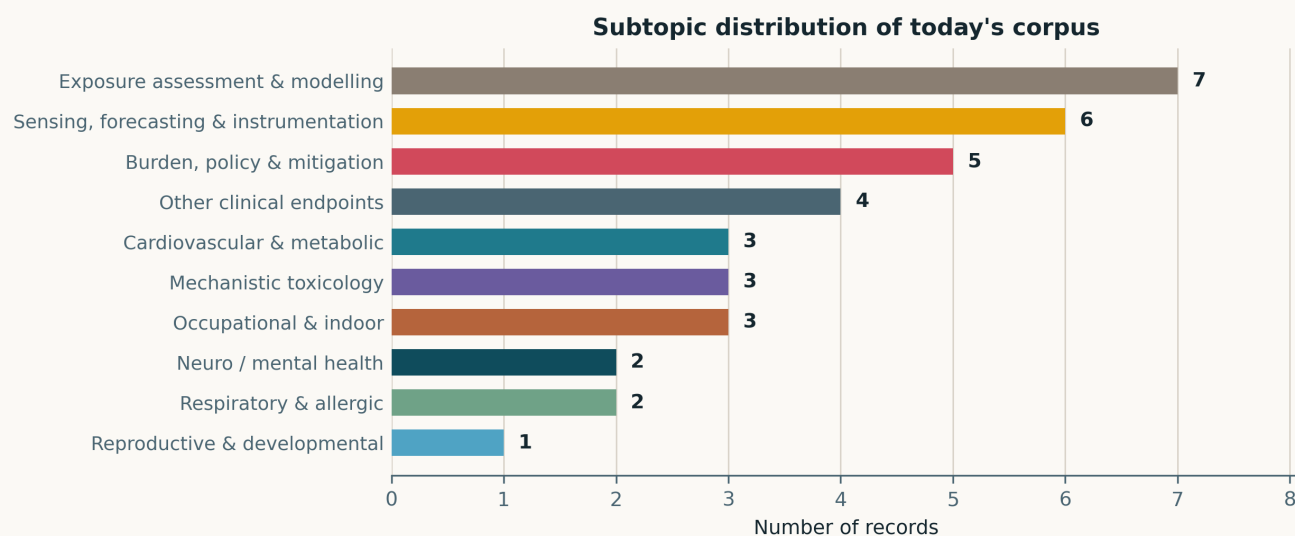


Fig. Records per subtopic cluster. All ten clusters are occupied — the fourth issue ever to manage it, and the first since 28 July. *Exposure assessment & modelling* 7 and *Sensing* 6 lead; *Reproductive & developmental* is the thinnest at 1. The completeness is a consequence of volume (36 records), not of breadth in the day's deposits: at 14 records this distribution would have left three or four clusters empty.

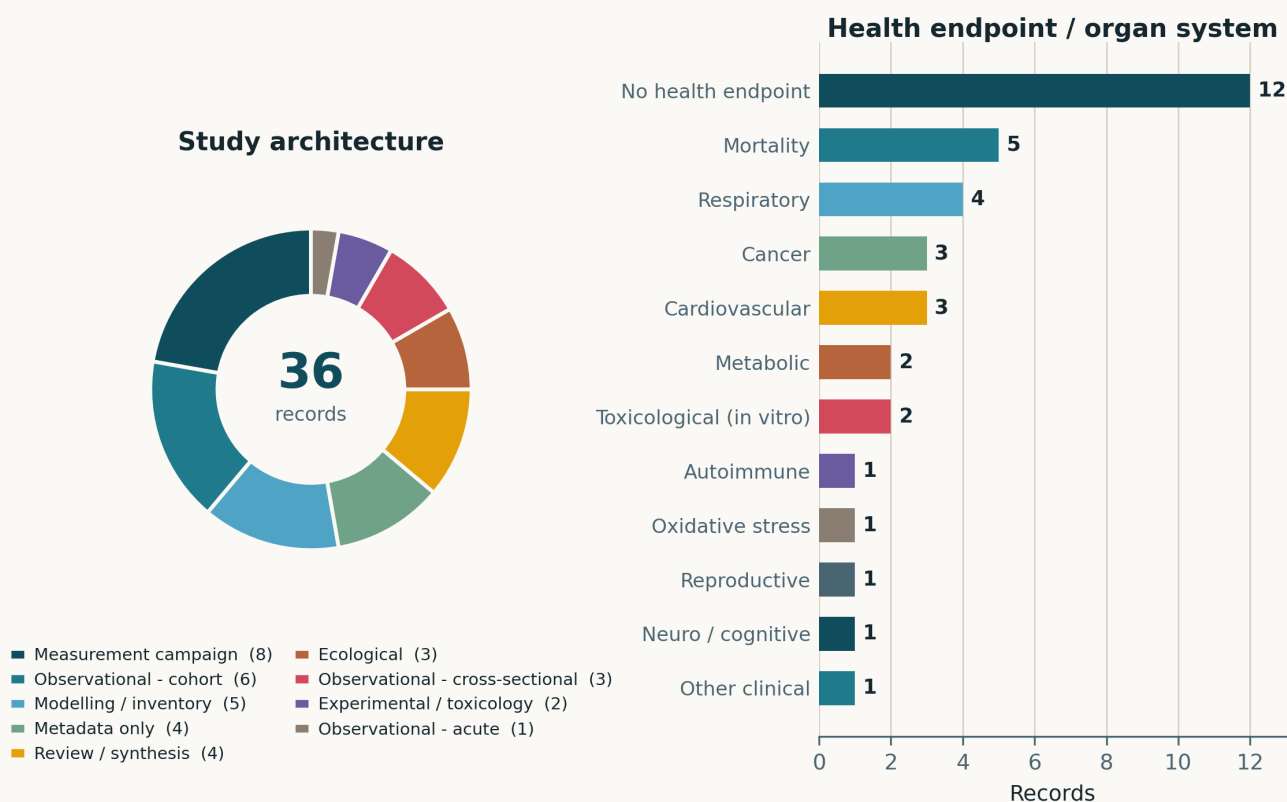


Fig. Left — study architecture. *Measurement campaign* 8, *Observational — cohort* 6, *Modelling / inventory* 5, *Review / synthesis* 4, *Metadata only* 4, *Observational — cross-sectional* 3, *Ecological* 3, *Experimental / toxicology* 2, *Observational — acute* 1. The *Metadata only* bar is a harvesting artefact, not a design: those four records have no abstract and their architecture is unknown. Right — health endpoint. **Twelve records carry no human health endpoint** (instrument or exposure work); *Mortality* 5 and *Respiratory* 4 lead the remainder, and seven endpoint categories are singletons.

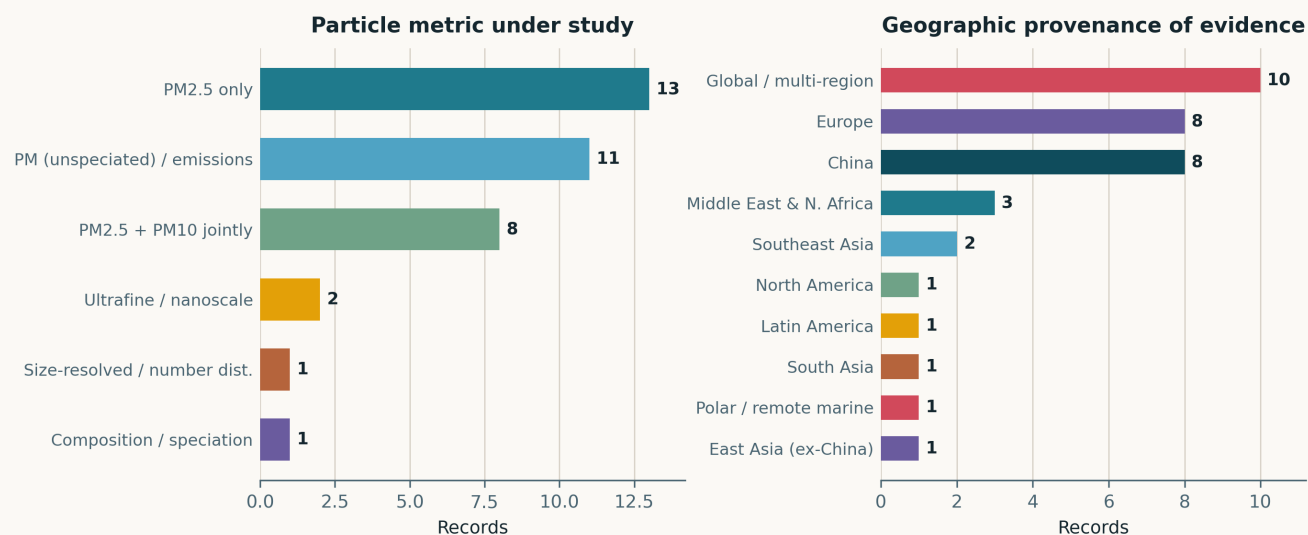


Fig. Left — particle metric. *PM_{2.5} only* 13, *PM (unspeciated) / emissions* 11, *PM_{2.5} + PM₁₀ jointly* 8, *ultrafine / nanoscale* 2, and one each for *composition / speciation* and *size-resolved / number*. The single *size-resolved* record is the PFAS worker panel, whose entire finding depends on the size cut. Note the grouper undercounts speciation: the Xi'an OP study, the New Mexico metals study and the Guangzhou toddler study all resolve constituents but bin as *PM_{2.5}-only*. Right — geography. **Global / multi-region 10 is inflated** — it absorbs both genuine multi-region syntheses and records whose site is not stated, including three of the four abstract-free deposits. China 8 and Europe 8 dominate the located records; North America contributes 1.

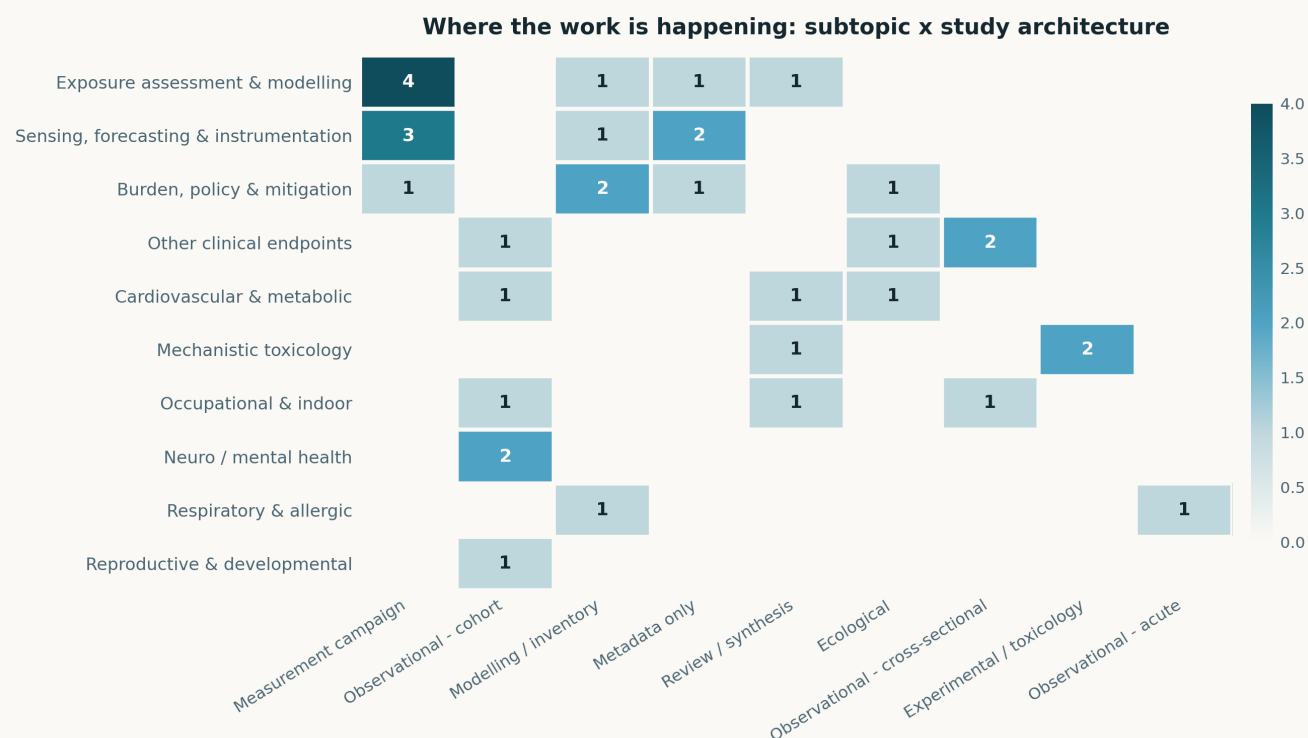


Fig. Subtopic × study-architecture cross-tabulation. The densest cell is *Exposure assessment* × *Measurement campaign* at 4, followed by *Sensing* × *Measurement campaign* at 3. **Five cells hold 2** — *Sensing* × *Metadata only*, *Burden* × *Modelling / inventory*, *Mechanistic toxicology* × *Experimental / toxicology*, *Neuro* × *Observational — cohort* and *Other clinical endpoints* × *Observational — cross-sectional* — and the remainder are singletons. The diagonal structure is the expected one — instrument and exposure work concentrates in measurement campaigns, clinical clusters in cohorts — and the only off-diagonal density is the pair of metadata-only records sitting in *Sensing*, which is an artefact of which publishers deposit abstracts, not of what was studied.

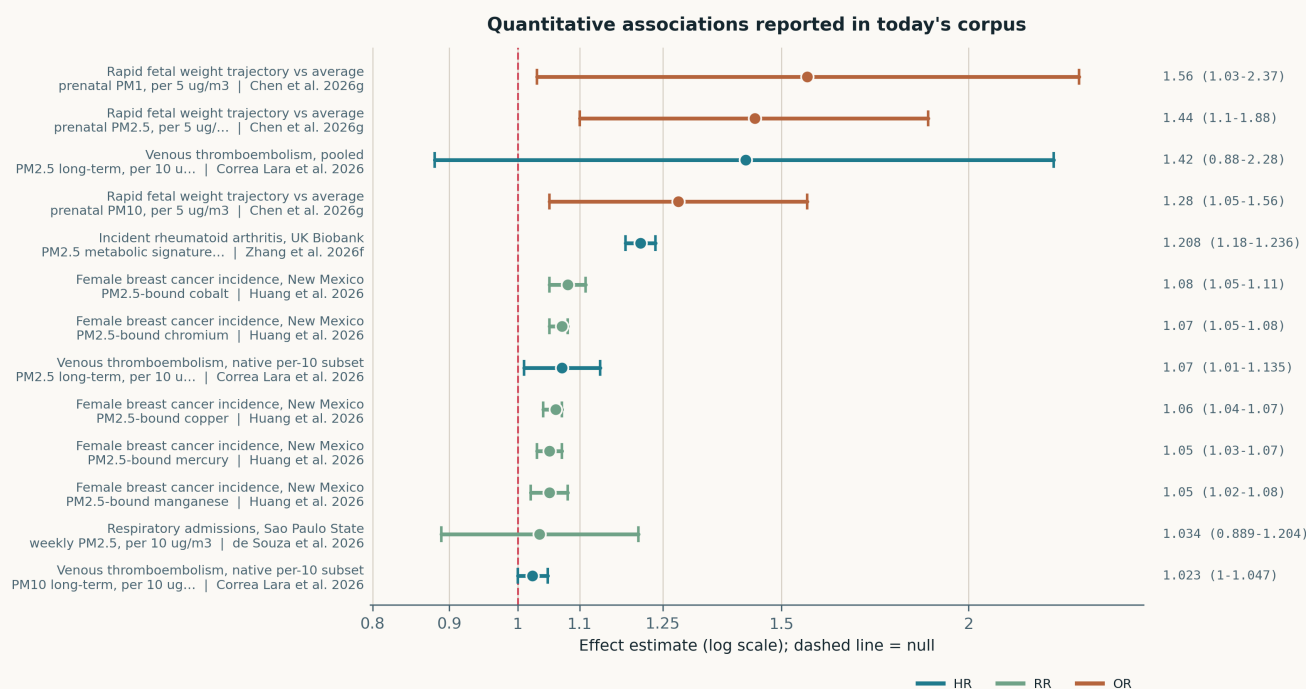


Fig. Thirteen interval-bearing estimates from five studies, all on particulate contrasts. **Three internally comparable sets:** the three VTE meta-analytic hazard ratios (same review, but the wide PM_{2.5} 1.42 estimate and the narrow 1.070 estimate are the *same exposure* under two harmonisation rules and must be read as a pair, not as two findings); the three per-5 $\mu\text{g}/\text{m}^3$ fetal-growth odds ratios, which are mutually comparable and ordered PM₁ > PM_{2.5} > PM₁₀; and the five New Mexico metal rate ratios, which share a cohort, an exposure model and an outcome. Two estimates are *not* exposure–outcome associations on a per-unit PM contrast: the UK Biobank RA hazard ratio is per SD of a *metabolic signature* derived from PM_{2.5}, not per unit PM_{2.5}, and belongs on this axis only in the weak sense that it shares a ratio scale. The São Paulo respiratory RR crosses the null (0.889–1.204) and its source reports $p = 0.054$; it is drawn because it is a per-10 $\mu\text{g}/\text{m}^3$ PM_{2.5} contrast, not because it is significant.

Life-course windows addressed today (records may span several stages)

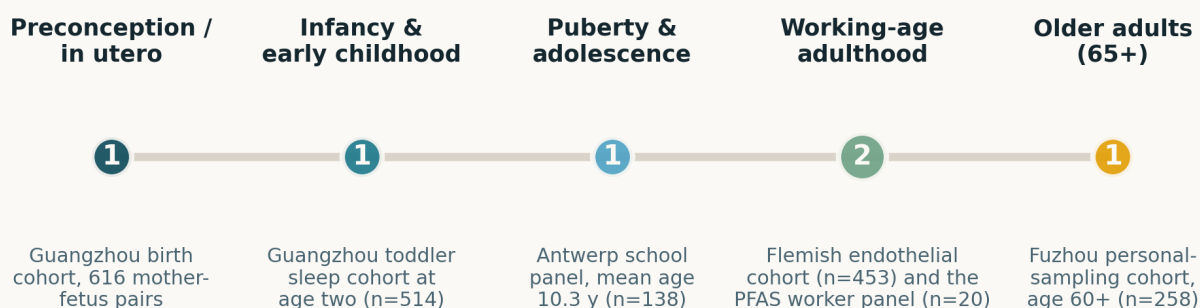


Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored to a stage. Six of 36 records qualify. **All five windows are occupied** — the first fully populated strip since 19 August, and only the seventh in this watch's history. The paediatric window breaks a run of four consecutive empty issues and the adolescent window a run of five. Working-age adulthood carries two (the Flemish endothelial cohort and the 20-worker PFAS panel); every other window carries exactly one, so the strip is coverage, not weight of evidence.

Paper-level digest

Ordered by cluster; the 27 August entry window only. Tier A/B/C is a triage of how far the record can be relied on, not of journal prestige.

Sensing, forecasting & instrumentation

6 records

Li et al. 2026c Plants (Basel)

Early warning of cucumber angular leaf spot from airborne pathogen aerosols estimated with PM sensors

Two-stage greenhouse surveillance in which **low-cost PM sensors act as continuous physical proxies** for suspended pathogen aerosol, and qPCR air sampling is triggered only when an estimated aerosol risk crosses a threshold. Evaluated on **732 plot-days** across spatially independent trials: an Extra Trees model predicted pathogen-aerosol accumulation at $R^2 = 0.884$, and an XGBoost model integrating aerosol load with microclimate forecast the daily disease-index change rate at $R^2 = 0.874$. SHAP identified aerosol concentration and vapour pressure deficit as the primary drivers. The limitation is stated by the authors and is the right one: PM sensors have no biological specificity, and the design compensates by scheduling a specific assay rather than by claiming the sensor sees pathogens. Cross-greenhouse rather than cross-site validation, and no reference-instrument comparison for the PM channel itself.

doi: 10.3390/plants15162510

Kang et al. 2026b J Hazard Mater

Emission characteristics and health impacts of vehicle-originated condensable PM

Condensable PM from **eight vehicle types** characterised with a purpose-built real-time system (PNU-CPM) plus offline physicochemical analysis, with emission factors derived from the real-time record. Untreated **diesel levels 4 and 5 emitted the most CPM and total PM**; CPM mass fractions were **63.2 % (diesel L4), 8.2 % (GDI) and 58.8 % (LPG 2)**. Organic CPM was ~53.5 % alkanes and ~5.1 % esters; dominant water-soluble ions were SO_4^{2-} and Na^+ . MPPD dosimetry gives higher deposited mass per body weight in younger age groups. Directly relevant to optical sensing: CPM is ultrafine, semi-volatile and forms after the tailpipe, so it is missed by hot-filter source testing and will not follow the scattering or hygroscopic-growth assumptions that low-cost sensor corrections inherit from combustion soot. No on-road fleet measurement — these are dynamometer emission factors.

doi: 10.1016/j.jhazmat.2026.143367

Lefevre et al. 2026 Data Brief

Urban microclimate dataset from low-cost sensor networks, Reunion Island

A data descriptor for **25 self-built LoRa weather stations** deployed on residential buildings across two contrasting neighbourhoods (Saint-Denis, Le Port), October 2024 to June 2025, at **10-minute cadence** for air temperature, relative humidity, wind speed, global horizontal solar radiation and grey-globe temperature. Station metadata, QC procedures and maintenance logs are published alongside, and incomplete stations are identifiable. **There is no PM channel** — this is a microclimate network, not an air-quality one — so its value here is as a deployment reference: dense neighbourhood-scale LoRa siting, self-built hardware, published QC, and an explicitly stated reuse path for model calibration (ENVI-met, STEVE). Tropical hot-humid deployments are exactly the regime in which PM sensor RH correction fails worst, and this is a rare open record of what such a network's data actually look like.

doi: 10.1016/j.dib.2026.113166

Wang et al. 2026d Atmos Chem Phys

Perchloric acid drives new particle formation enhanced by dimethylamine, ammonia and sulfuric acid

Quantum-chemical calculations coupled to the Atmospheric Clusters Dynamic Code compare the nucleation enhancement of DMA, NH_3 and H_2SO_4 for perchloric-acid-based new particle formation, motivated by recent Arctic observations of atmospheric HClO_4 . Both DMA and NH_3 bind PA strongly through hydrogen bonding and proton transfer, and **PA nucleates more readily with DMA than sulfuric acid does**; the PA-DMA cluster formation rate exceeds PA- NH_3 even at overwhelming NH_3 excess, with 1:1 PA:DMA clusters controlling growth. Purely computational — no chamber or field verification, and the claimed NH_3 excess of “10–100 orders of magnitude” is a transcription problem in the abstract rather than a physical statement. Relevant as a candidate nucleation channel in clean marine and polar boundary layers where sulfuric acid is scarce, which is where number-concentration instruments see unexplained events.

doi: 10.5194/acp-26-12171-2026

Moteki et al. 2026 J Aerosol Sci

Refractive index of fractal-like aggregates from complex forward-scattering amplitudes

Metadata-only deposit — title and DOI without an abstract. The method named in the title is the single-particle complex forward-scattering amplitude $S(0^\circ)$ approach, previously used to constrain the black-carbon refractive index at $\lambda = 0.633 \mu\text{m}$; the new element is **mass classification of the aggregates** before the optical measurement, which removes the size-distribution term that dominates the uncertainty in bulk retrievals. If it delivers, this is the most directly useful sensing record in the window: every optical PM instrument, from a nephelometer to a \$20 laser scatter sensor, converts scattered intensity to mass through an assumed refractive index, and for soot-dominated aerosol that assumption is the largest single error term. Carried at tier C on the title alone and placed at the top of the abstract retry list.

doi: 10.1016/j.jaerosci.2026.106884

Chen et al. 2026e Atmos Environ

Aerosol and VOC pollution characteristics along the lower Yangtze, from shipborne observation

Metadata-only deposit. Shipborne mobile measurement of aerosol and VOCs along an inland waterway corridor — a platform class this watch sees rarely and one with a specific methodological problem: the measurement frame moves through the plume it is sampling, so concentrations are Lagrangian-contaminated in a way that fixed-site data are not, and self-sampling of the vessel's own exhaust is a standing artefact that requires wind-relative rejection to control. Whether the authors address either is not determinable from the title. Carried at tier C; the record is retained rather than rejected because inland-shipping emissions are a genuinely under-monitored PM source and the corridor is one of the most heavily trafficked in the world.

doi: 10.1016/j.atmosenv.2026.122331

Exposure assessment & modelling

7 records

Rashid et al. 2026 Toxics*Ultrafine particles and mortality: scoping review of evidence and exposure-assessment limits*

PRISMA-ScR scoping review of PubMed and Scopus: **21 of 704 screened articles**, 2007–2025, split **14 short-term** (time-series or case-crossover against central-site particle number counters) and **7 long-term** (land-use regression, chemical transport or machine-learning surfaces). **10 of 14 short-term and 6 of 7 long-term studies reported positive associations**, strongest for respiratory mortality. The finding that matters for this watch is the second-order one: **the strongest associations came from the studies with higher-resolution or source-specific exposure assessment**, a pattern consistent with reduced Berkson-type misclassification rather than with true effect heterogeneity. A scoping review does not pool, so there is no summary estimate and no formal heterogeneity statistic; the exposure-method gradient is observed, not tested. Still the clearest statement to date of why UFP regulation stalls on measurement.

doi: 10.3390/toxics14080664

Chen et al. 2026f Toxics*Source-specific oxidative potential of PM_{2.5}: water-soluble metals via DTT and interpretable ML*

Winter and summer ambient PM_{2.5} in Xi'an assayed for DTT oxidative potential, with XGBoost–SHAP attribution over composition and receptor-model source contributions. **Volume-normalised OP (DTT_v) was markedly higher in winter**, and although water-soluble metals are a small fraction of PM_{2.5} mass, **potassium and manganese dominated the OP attribution** — a biomass-burning tracer and a redox-active transition metal. Source apportionment separates the two seasons cleanly: dust drives winter OP through mass loading, traffic drives summer OP through intrinsic per-mass toxicity. The abstract reports no numeric DTT_v values, confidence intervals or model performance, which is a real gap for a paper whose central claim is quantitative attribution; SHAP importance is also not a causal decomposition and correlated constituents will trade rank. The qualitative conclusion — OP tracks composition and source, not mass — is the second independent instance of it this week.

doi: 10.3390/toxics14080646

Li et al. 2026d Environ Pollut*Surface PM transport under a closed earth-air model*

One year of continuous in-situ monitoring in the semi-enclosed basement of the Mogao Grottoes conservation department, testing whether **earth-air vertical movement** driven by atmospheric-pressure fluctuation controls surface PM independently of wind and human activity. Structural equation modelling gives **CO₂ as a tracer for EVM-driven PM dynamics**, with path coefficients of **0.452 (PM_{2.5})**, **0.470 (PM₁)** and **0.386 (PM₁₀₀)**; ²²²Rn regulates PM positively but far more weakly, and relative humidity is strongly inhibitory. PM_{2.5} and PM₁ are strongly affected by EVM, PM₁₀₀ only weakly. Single site, single enclosure, and SEM path coefficients on autocorrelated one-year time series overstate precision badly. The mechanism claim is interesting for indoor and heritage monitoring; the effect sizes should not be transported.

doi: 10.1016/j.envpol.2026.128983

Mainka & Mainka 2026 Preprints.org*Urban cycling versus PM_{2.5} exposure: does the benefit outweigh the risk?*

Scenario analysis of **162 real commuting trips by two cyclists** in Gliwice across heating and non-heating seasons, with personal PM_{2.5} measured by low-cost sensor and inhaled dose reconstructed from concentration, trip duration and cycling-specific ventilation. Median trip 33.0 min/day. The median physical-activity relative risk was **0.836** (a 16.4 % all-cause mortality reduction) against a median PM_{2.5} relative risk of **1.00182** (0.182 % increase), for a **net RR of 0.840 (IQR 0.815–0.879)**; **160 of 162 trips retained net benefit** under sensitivity analysis. Two hard limits: *n* = 2 **commuters**, so the exposure distribution is a convenience sample of two routes, and the risk functions are transported from published cohorts rather than estimated. Unreviewed preprint. The useful part is the method — trip-resolved dose from a low-cost sensor is the right exposure object for this question, and almost no one computes it.

doi: 10.20944/preprints202608.1778.v1

Moryani et al. 2026 Toxics*Bioaccessibility and speciation of heavy metals in PM_{2.5} road dust*

48 grid-based samples across four functional zones (residential, school, hospital, office) in Karachi (>27 million) and Shikarpur (~0.3 million), with the PM_{2.5} fraction of road dust extracted in artificial lysosomal fluid (**pH 4.5, 37 °C, 24 h**) and speciation modelled in PHREEQC and Visual MINTEQ. **Cr, As and Pb showed moderate inhalation bioaccessibility**; contamination and ecological-risk indices were computed alongside. The design addresses a real gap — total metal content is a poor predictor of the dose that reaches lung tissue — but ALF extraction is a simulation with no in-vivo anchor, and the abstract reports no bioaccessible fractions, no concentrations and no zone contrasts, so nothing quantitative can be lifted. Road dust is also resuspended, not directly emitted, which the pollution indices do not distinguish.

doi: 10.3390/toxics14080704

Isworo & Oetari 2026 Environ Sci Pollut Res Int*Risk-integrated industrial air quality: Gaussian dispersion linked to a health-weighted pressure index*

Gaussian-plume dispersion for a Central Java textile facility combined with a hazard-quotient-normalised Atmospheric Environmental Pressure Index, using an emission inventory, ERA5 2015–2024 meteorology and six receptor measurements. The headline is a compliance paradox worth stating: **modelled concentrations meet Indonesia's national standards while PM_{2.5} exceeds the WHO 2021 guideline**, and PM_{2.5} and PM₁₀ dominate the index. AEPI values (0.06 baseline, 0.08 predicted) sit in the authors' own "very low" band, which makes the index insensitive at exactly the concentrations where the health argument bites — a scaling problem the paper does not confront. One facility, six receptors, no validation of the plume model against the measurements. Tier C: the framework is transferable, the result is not evidence.

doi: 10.1007/s11356-026-38159-z

Wu et al. 2026b npj Clim Atmos Sci*Global fire activity exacerbates extreme PM_{2.5} pollution*

Metadata-only deposit. The title claim — that fire activity acts disproportionately on the *extreme tail* of the PM_{2.5} distribution rather than on the mean — is the one that matters, because attainment statistics, exposure-response functions fitted on annual means and most burden calculations are all mean-anchored and will understate a tail-loading source. It is also the regime in which low-cost networks are most useful and least characterised: sensor response at several hundred $\mu\text{g m}^{-3}$ of smoke is outside the concentration range of essentially every published co-location study. Carried at tier C pending the abstract; a Nature-portfolio deposit with no abstract in Crossref is unusual and may resolve on retry.

doi: 10.1038/s41612-026-01527-0

Correa Lara et al. 2026 Thromb Haemost*Ambient PM and venous thromboembolism: updated systematic review and meta-analysis*

19 observational studies, >4.5 million participants, searched across PubMed, Embase, Web of Science and Cochrane, harmonised to $10 \mu\text{g}/\text{m}^3$ and pooled under restricted maximum likelihood with Hartung–Knapp adjustment. Overall long-term: **PM_{2.5} HR 1.42 (0.88–2.28)**, $I^2 = 97.0\%$; **PM₁₀ HR 1.18 (0.85–1.64)**, $I^2 = 96.2\%$. Restricting to studies natively reporting per $10 \mu\text{g}/\text{m}^3$: **PM_{2.5} 1.070 (1.010–1.135)** and **PM₁₀ 1.023 (1.000–1.047)**, both at $I^2 = 0\%$. Short-term exposures were null. The sensitivity analysis is the finding — near-total heterogeneity removed by excluding rescaled contrasts — but it is also a smaller, non-random subset, and a subset selected on a reporting characteristic can be homogeneous for reasons other than correctness. Inconsistent VTE case definitions across the 19 studies remain unaddressed.

doi: 10.1055/a-2945-4974

Yang et al. 2026b Blood Press*Endothelial function at low-level chronic residential air pollution*

453 Flemish adults (47.7% women, mean age 52.8 y) with endothelial function by finger photoplethysmography after 5 min ischaemia, related to residential PM₁₀, PM_{2.5}, NO₂ and black carbon from high-resolution spatiotemporal interpolation. Six-year means were **15.9, 12.8, 14.3 and 1.04 $\mu\text{g}/\text{m}^3$** — at or below current EU limits and near the WHO interim range. **Dmax was inversely associated with PM_{2.5}** irrespective of risk-factor adjustment; other pollutant–outcome pairs were consistently inverse but weaker ($p < 0.10$). Association sizes **weakened as pollution fell over 2010–2015** ($p \leq 0.044$), and the spatial gradient in endothelial dysfunction tracked the PM_{2.5} gradient. Both endothelial measures predicted a composite cardiovascular endpoint (HR 0.27–0.49). Modest n , and no confidence intervals are given for the exposure–endothelium associations themselves, which is why none appear in the forest plot. The attenuation-with-decline result is the notable one and is rarely reported.

doi: 10.1080/08037051.2026.2724736

Ramírez-Moreno et al. 2026 Med Sci (Basel)*Regional stroke hospitalisation and mortality against long-term PM_{2.5}*

Nationwide ecological analysis of Spain's 17 autonomous communities, 2013–2021, **306 sex-specific region-year observations**, linking INCLASNS stroke indicators to MITECO ambient pollution. Mean age-standardised hospitalisation rate 20.1 ± 5.4 per 10,000. **PM_{2.5} had the largest positive correlation with age-standardised mortality of any pollutant examined** ($r = 0.264$, $p < 0.001$), which the authors correctly describe as weak; associations with hospitalisation and with other pollutants were small and inconsistent in sign. **Regional variation in stroke burden was not explained by the available pollution indicators**. Exposure is at region-year resolution against sex-stratified outcomes, so the sex analysis is uninterpretable by construction and the authors say so. Reported here as a well-labelled null; the design cannot support individual-level inference and does not claim to.

doi: 10.3390/medsci14040475

Neuro / mental health**2 records****Wu et al. 2026c** Toxics*Early-life PM exposure and toddler sleep: particle size and constituents*

Birth cohort of **514 mother–child pairs** in Guangzhou followed to age two (2017–2018), with PM₁, PM_{2.5}, PM₁₀ and PM_{2.5} constituents from high-resolution ground-level grids and sleep by the Chinese Brief Infant Sleep Questionnaire. Per-IQR early-life exposure was associated with reduced nighttime sleep duration: **PM₁ $\beta = -0.18$ (95% CI $-0.32, -0.04$)**, **PM_{2.5} -0.23 ($-0.38, -0.08$)**, **PM₁₀ -0.21 ($-0.35, -0.07$)** hours. Quantile g-computation on the constituent mixture reproduced the decrement with **SO₄²⁻ as the dominant contributor**, and pregnancy-window findings were consistent. Outcome is parent-reported, which biases toward non-differential misclassification but not necessarily away from confounding by household environment; the three particle sizes are collinear at this resolution, so the size ordering is suggestive rather than discriminating. Sulfate as the driver is consistent with the secondary-inorganic signal this watch carried on 26 August from an unrelated endpoint.

doi: 10.3390/toxics14080653

Hendrickx et al. 2026 Int J Environ Res Public Health*Recent air pollution exposure, children's neurocognition and endothelial function*

138 healthy children (mean age 10.3 y, 55.8% male) across seven Antwerp primary schools including one open-air school, with indoor and outdoor PM and outdoor NO₂ measured and endothelial and attentional outcomes assessed at three repeated visits. Exposure was resolved at **4 h (direct), 24 h (acute) and 1 week (recent) windows**. **Higher direct indoor PM was associated with reduced microvascular dilatation**, while **selective and sustained attention improved with direct and recent PM exposure** respectively — a sign reversal the authors report without resolving, and which most plausibly reflects the same seasonal or activity pattern driving both classroom occupancy and outdoor concentration rather than a cognitive benefit. No NO₂ associations. Excluding the open-air school shifted results, which on $n = 138$ with seven schools means the school-level clustering is doing real work. Tier B for the vascular finding; the attention result should not be cited without the full text.

doi: 10.3390/ijerph23081070

Respiratory & allergic**2 records****Morvan et al. 2026** Int J Epidemiol*Causal effect of PM_{2.5} on COVID-19 hospitalisation and death, conditioned on comorbidity*

Instrumental-variable analysis over **more than 3.5 million weekly municipality observations** covering all of France, 2020–2021, with COVID-19 hospitalisations and deaths disaggregated by age, sex and municipality and linked to records of pre-existing conditions. **Wind direction instruments local PM_{2.5} in a two-stage least squares design**. Among individuals *without* pre-existing conditions there was **no significant effect** on either outcome. Conditioning on comorbidity, a $10 \mu\text{g}/\text{m}^3$ increase was associated with **+1.5 COVID-19 deaths per 100,000 (95% CI 0.8–2.2)**, concentrated in respiratory (**0.9, 0.4–1.4**) and cardiovascular (**0.6, 0.1–1.1**) subgroups, and driven by the oldest age strata. Wind direction satisfies exclusion only if it affects COVID outcomes solely through PM_{2.5} — doubtful during a period when weather drove both mobility and indoor crowding — and the estimates are ecological rate differences, not individual risks. Still the strongest identification strategy this literature has produced.

doi: 10.1093/ije/dyag152

de Souza et al. 2026 Sci Total Environ

Wildfire smoke, PM_{2.5} and respiratory morbidity: spatial mapping and time series

1,467,172 respiratory hospital admissions in São Paulo State, 2018–2025 (DATASUS), against **20,446 INPE-recorded wildfire events** and daily CETESB PM_{2.5}, temperature and humidity, analysed by GAM on weekly aggregates with adjustment for meteorology, calendar month and a nonlinear trend. Children 0–10 and males carried the greatest burden; population-standardised rates were highest in the north and north-west, coinciding with fire hotspots, while absolute counts concentrated in the metropolitan area — a textbook illustration of why per-capita and count maps disagree. **Each 10 µg/m³ weekly PM_{2.5} was associated with a 3.4 % increase in admission rates (RR 1.034, 95 % CI 0.889–1.204, $p = 0.054$), strongest at zero lag.** Fire counts and PM_{2.5} correlated at $\rho = 0.732$. The GAM explains 82.8 % of deviance, but weekly aggregation of an acute exposure throws away most of the contrast and is the likeliest reason the interval spans the null.

doi: 10.1016/j.scitotenv.2026.182227

Reproductive & developmental

1 record

Chen et al. 2026g Toxics

Prenatal size-fractionated PM and fetal growth trajectories, mediated by maternal glucose

616 mother–fetus pairs in Guangzhou with repeated mid-to-late-pregnancy ultrasound (HC, AC, FL, BPD as z-scores; EFW trajectories classed average, slow or rapid) and residential PM₁, PM_{2.5}, PM₁₀ from 1 km spatiotemporal models. Each 5 µg/m³ increase raised the odds of a *rapid* EFW trajectory versus average: **PM₁ OR 1.56 (1.03–2.37), PM_{2.5} 1.44 (1.10–1.88), PM₁₀ 1.28 (1.05–1.56)** — monotone in the inverse of particle size, as a deposition argument would predict. BPD, HC and AC z-scores moved in the same direction. **Maternal fasting glucose and insulin resistance mediated 5.0–29.4 %** of the association. The mediation range is wide enough to be uninformative at its lower end, mid-pregnancy glucose is a single measurement, and *accelerated* fetal growth is a less established adverse outcome than restriction, so the clinical framing is the weakest part. The size gradient is the citable result.

doi: 10.3390/toxics14080709

Mechanistic toxicology

3 records

Qin et al. 2026b Environ Health (Wash)

Engineered protein coronas: ApoE enhances particulate transport across the blood-brain barrier

Gold nanoparticles functionalised with thiolated DNA aptamers serve as a **compositionally controlled PM surrogate**, allowing reproducible assembly of coronas from three physiologically relevant proteins — human serum albumin, apolipoprotein E and complement C3b — and isolating corona identity from particle identity. In a bEnd.3 Transwell BBB model, **ApoE-enriched coronas nearly doubled apparent permeability and increased 24 h transport 1.6-fold** over bare particles, through an energy-dependent, predominantly clathrin-mediated route. ApoE precoronation also raised endothelial penetration of polystyrene particles and **increased the endothelial cytotoxicity of real industrial-source PM samples**, which is the step that connects the model to ambient aerosol. The surrogate is the limitation: gold nanoparticles share neither the surface chemistry nor the morphology of combustion PM, and a Transwell monolayer is not a neurovascular unit. Mechanistically the cleanest account yet of why particle *surface*, not particle mass, governs neural access.

doi: 10.1021/envhealth.5c00803

Moonwiriakit et al. 2026 J Vis Exp

A STAT6 reporter assay for screening toxicants affecting allergic airway inflammation

A methods paper establishing a stable STAT6-responsive luciferase reporter in **16HBE14o- human bronchial epithelial cells** by lentiviral transduction, with puromycin selection, MOI optimisation and monoclonal isolation by limiting dilution. Responsiveness was validated against IL-4/IL-13, pathway specificity confirmed with selective STAT6, STAT5 and STAT3 inhibitors, and reproducibility quantified by **Z'-factor**. Applied to individual air-pollution constituents, **benzo[b]fluoranthene and particulate matter both significantly increased STAT6 reporter activity**. No dose–response, no PM characterisation (source, size, mass loading) and no Z' value are given in the abstract, so the assay's screening sensitivity cannot be assessed from it. The value is infrastructural: allergic-airway signalling is a plausible route for PM-associated asthma incidence and there has been no scalable prioritisation assay for it.

doi: 10.3791/71694

Nikaido-Landry et al. 2026 Philos Trans R Soc Lond B Biol Sci

EGFR dysregulation and climate change: pollution-promoted lung cancer

Discussion-meeting review connecting rising anthropogenic fine PM to lung cancer in never-smokers via EGFR signalling. The mechanistic core is the promotion model: **fine PM promotes malignant transformation in cells that already carry an EGFR mutation**, so the relevant population is people with oncogenic mutations in histologically normal lung tissue rather than people accumulating new mutations from exposure. That reframing matters — it predicts a susceptible subgroup rather than a population-wide dose–response, and it implies precision-prevention strategies targeted by genotype rather than universal exposure reduction. A narrative review with no systematic search and no new data; the promotion model rests principally on one line of murine and cohort work, and the “climate change” framing in the title is rhetorical rather than analytic. Tier B for the mechanistic argument.

doi: 10.1098/rstb.2024.0508

Occupational & indoor

3 records

Li & Zheng 2026 Toxics

Size-resolved PM₁₀-bound PFAS: respiratory deposition and oxidative stress in recycling workers

Panel study at a southern-Chinese waste recycling plant: **15 size-resolved PM samples and 280 repeated first-void urines from 20 workers over 45 consecutive days**, with linear mixed models linking size-resolved PM-bound PFAS to urinary PFAS and to 8-OHdG and MDA. Short-chain PFAS predominated in both matrices; urinary ICCs were fair to excellent (>0.4). **PFAS in PM_{2.1–10} correlated more strongly with internal dose than PFAS in PM_{2.1}** — the opposite of the fine-fraction-dominates prior, and consistent with mucociliary clearance and gastrointestinal uptake of coarse-fraction deposits. **Each unit increase in ln average daily intake was associated with 10.2–12.6 % higher 8-OHdG (PM_{4.7–5.8}, PM_{9.0–10}) and 8.0–14.6 % higher MDA (PM_{5.8–9.0})**. Twenty workers, one plant, and the authors call the biomarker associations suggestive; with this many size bins the multiplicity is substantial and no correction is described. The size-resolved design is nonetheless the right one and is almost never done.

doi: 10.3390/toxics14080682

Sadki et al. 2026 *Cad Saude Publica*

Real-time formaldehyde exposure and acute symptoms in healthcare workers

Cross-sectional study of **180 healthcare workers across 12 hospital departments** in Morocco, with airborne formaldehyde measured over full eight-hour shifts by calibrated semiconductor dosimeter alongside a structured symptom questionnaire. **Every mean concentration was below the Moroccan occupational limit of 0.25 mg/m³** (0.048–0.196 mg/m³, 19–78 % of the limit), yet highly exposed workers reported markedly more eye irritation (72.5 %), headache (63 %) and respiratory discomfort (56.5 %). **Prolonged exposure (≥ 8 h/day, ≥ 3 days/week) independently predicted multiple symptoms (AOR 3.2, 95 % CI 1.7–5.9)**, as did inadequate ventilation (2.4) and inconsistent PPE use (1.9); years of exposure correlated with symptom count ($\rho = 0.41$). **The exposure is gaseous, not particulate**, so the AORs are excluded from this issue's forest plot. Carried because the deployment pattern — low-cost real-time dosimetry as routine occupational surveillance in a resource-limited setting — is exactly the use case for distributed sensing, and because symptom self-report against measured exposure is open to reporting bias in the obvious direction.

doi: 10.1590/0102-311XEN191325

Perna et al. 2026 *Life (Basel)*

Indoor air pollution as a determinant of airport occupational exposure

Narrative review arguing that airport occupational exposure assessment has been anchored on aircraft engine emissions and has under-weighted **indoor air quality driven by indoor-outdoor exchange and ventilation**. Ultrafine particles, black carbon, PAHs and trace metals are identified as the constituents driving oxidative stress, inflammation and xenobiotic pathway activation, with effects framed beyond the respiratory system. The operationally useful recommendation is to monitor indoor microclimatic parameters — temperature, pressure, humidity — as **event detectors for indoor air-quality deterioration**, which is a cheap addition to any fixed deployment. No systematic search, no quantitative synthesis, no exposure or effect estimates; the closing suggestion that AI-based exposure modelling is promising is asserted with the authors' own caveat that it lacks validation. Tier C: useful as a framing document for airport deployments, not as evidence.

doi: 10.3390/Life16081242

Burden, policy & mitigation

5 records

Özmen et al. 2026 *Toxics*

Long-term PM_{2.5} and premature mortality across Türkiye

Province-level cross-section of **all 81 Turkish provinces** for 2022, combining annual mean PM_{2.5} with WHO AirQ+ attributable-death estimates for adults 30+ at a 5 µg/m³ counterfactual. **Population-weighted mean PM_{2.5} was 27.0 µg/m³ — 5.4× the WHO guideline — and all 81 provinces exceeded it**. Bivariate OLS explained 41 % of between-province variation in attributable mortality (**slope 3.23 additional deaths per 100,000 per 1 µg/m³, 95 % CI 2.37–4.10**); a log-log model gave an elasticity of **0.80 (0.65–0.95)**. The attributable fraction of natural-cause mortality rose from **8.8 % in the lowest exposure quartile to 24.6 % in the highest, totalling 68,440 premature deaths, 14.2 % of natural-cause mortality**. The authors are explicit about the circularity and it is worth repeating: attributable deaths are *generated* by AirQ+ from a fixed concentration–response function, so regressing them on concentration recovers the assumed function, not independent evidence. The burden magnitude stands; the gradient is arithmetic.

doi: 10.3390/toxics14080728

Sommar et al. 2026 *Int J Environ Res Public Health*

Health economics of PM_{2.5} from residential wood combustion, northern Sweden

Health impact assessment for **280,742 residents of Västerbotten County** using high-resolution 2023 dispersion-modelled PM_{2.5} from small-scale residential heating, population-weighted to home-address coordinates, with the VALESOR framework and a near-source relative risk of **1.26 per 10 µg/m³** for natural-cause mortality. **Mean RWC-attributable exposure was only 0.16 µg/m³ (IQR 0.10–0.21)**, yielding **30 years of life lost annually** plus small morbidity counts (0.3 lung cancers, 4 childhood asthma, 3 COPD, 3 IHD, 1 stroke, 2 dementia) and a total annual cost of **~EUR 7.0 million (EUR 4.6–9.4 million)**. Two assumptions carry the result: the near-source RR of 1.26 is far above the standard ~1.08 per 10 µg/m³ and is a deliberate wood-smoke-toxicity premium that the evidence base supports only weakly, and the population-mean of 0.16 µg/m³ conceals a highly skewed near-source distribution. Fractional case counts at this precision are not meaningful.

doi: 10.3390/ijerph23081053

Zhai et al. 2026 *Front Public Health*

Global disparities in lifetime lung cancer risk: development, smoking and fine PM

GLOBOCAN 2022 with the adjusted-for-multiple-primaries method, giving **global lifetime risk of lung cancer incidence 4.91 % and mortality 4.07 %**, ranging from North America (7.30 % / 5.38 %) to Africa (0.99 % / 0.93 %) and higher in males on every continent. In country-level regression, smoking prevalence predicted both (LRI $\beta = 0.001$, $p < 0.001$), and **PM₁ was the strongest environmental predictor (LRI $\beta = 0.40$, $p = 0.001$; LRM $\beta = 0.20$, $p = 0.006$)**, slightly ahead of PM_{2.5} (0.36 and 0.18). Higher national development correlated with higher lifetime risk. The ecological confounding is total: HDI, smoking history, registry completeness and diagnostic capacity all covary with PM at country level, and lung cancer incidence in high-HDI countries is dominated by historical smoking and by the ability to detect it. **PM₁ outranking PM_{2.5} is the one result worth carrying**, and even that is weak given how few countries have independent PM₁ products.

doi: 10.3389/fpubh.2026.1842037

Acat et al. 2026 *Atmosphere*

National trends in ambient pollutants and cause-specific mortality, Türkiye 2018–2024

Seven annual national observations for PM₁₀, SO₂, CO, NO₂ and O₃, each the unweighted mean of eligible station annual means from official bulletins, against crude national mortality rates, tested by OLS with Mann–Kendall, Sen's slope and a strict common-station sensitivity analysis. **PM₁₀ fell from 50.4 to 41.0 µg/m³** and SO₂ from 13.2 to 8.5, both significant. The all-station NO₂ indicator rose from 29.2 to 34.4 but **the trend vanished under common-station restriction (slope −0.021, $p = 0.950$)** — a clean demonstration that a national mean over a changing monitoring network measures the network as much as the air. **PM_{2.5} was excluded outright** for insufficient reporting comparability, which is itself the story for a country whose modelled PM_{2.5} burden the AirQ+ record above puts at 68,440 deaths. Only neoplasm mortality declined significantly. Crude national rates against unweighted station means; no causal reading is possible and none is claimed.

doi: 10.3390/atmos17090832

Zhu et al. 2026 Atmos Pollut Res*Short-term effectiveness of urban road cleaning for fugitive dust, from TRAKER mobile monitoring*

Metadata-only deposit. TRAKER (Testing Re-entrained Aerosol Kinetic Emissions from Roads) infers road-dust emission potential from the difference between behind-tyre and background aerosol measured on a moving vehicle, which makes it one of the few mobile methods that measures an *emission* rather than a concentration. The question in the title — how long the benefit of a street-sweeping pass actually persists — is one municipalities repeatedly fund cleaning programmes without answering, and the honest prior from the existing literature is “hours to a few days, if at all”. No abstract, no site, no numbers. Carried at tier C and retained because the method class is directly relevant to mobile PM sensing.

doi: 10.1016/j.apr.2026.103190

Other clinical endpoints

4 records

Huang et al. 2026 Air Qual Atmos Health*PM_{2.5}-bound metals and female breast cancer incidence, New Mexico*

Census-tract analysis of New Mexico Tumor Registry breast-cancer incidence, 1997–2019, against annual PM_{2.5}-bound metal exposure intensities from a calibrated Emission Weighted Proximity Model over a **10-year pre-diagnosis window**, fitted by GEE with Gaussian estimation. All five metals were associated with higher age-adjusted incidence: **chromium RR 1.07 (1.05–1.08)**, **cobalt 1.08 (1.05–1.11)**, **copper 1.06 (1.04–1.07)**, **manganese 1.05 (1.02–1.08)**, **mercury 1.05 (1.03–1.07)**, with a monotone trend across unexposed/low/medium/high categories for every metal. The intervals are strikingly narrow, which is the tell: these are tract-level rate ratios on 23 years of registry data, so precision reflects the number of tract-years rather than the number of informative exposure contrasts, and the five metals are emitted by shared sources and are almost certainly collinear — five nominally independent estimates from one exposure surface. EWPM is a proximity model, not a measurement. Ecological throughout.

doi: 10.1007/s11869-026-02079-1

Zhang et al. 2026f Front Public Health*Plasma metabolomic signatures of air pollution and rheumatoid arthritis risk*

403,332 RA-free UK Biobank participants followed a median **14.01 years**, with residential annual PM_{2.5}, PM₁₀, NO₂, NO_x and a composite air pollution score, plasma metabolomics, machine-learning-derived pollutant-specific metabolic signatures, Cox models, generalised propensity score balancing and mediation analysis. Per SD of signature: **PM_{2.5}-related HR 1.208 (1.180–1.236)**, NO₂ 1.141, NO_x 1.174, APS 1.184, stable under propensity balancing. **Signatures accounted for 15.7–21.1 %** of the pollution–RA associations. The critical reading: the hazard ratios are for the *signature*, a metabolite score trained to predict exposure, not for exposure itself — a signature can outperform its own target because it absorbs correlates of exposure (diet, adiposity, smoking) that the pollutant variable does not. Single-timepoint metabolomics, UK Biobank’s healthy-volunteer selection, and a mediation framework that assumes no exposure–mediator confounding.

doi: 10.3389/fpubh.2026.1901951

Chen et al. 2026h Metabolites*Metabolic perturbation in elderly adults exposed to PM_{2.5}-bound ester compounds*

258 Fuzhou residents aged 60+ with 72-hour personal PM_{2.5} sampling (UPAS V2), GC-MS speciation and LC-MS plasma metabolomics, targeting non-phthalate, non-organophosphate esters that routine speciation ignores. Mean personal PM_{2.5} was **38.06 µg/m³** with esters dominating the measured organic composition; 20 high-concentration esters were prioritised. Covariate-adjusted linear regression associated ester exposure with **steroid biosynthesis, glycolysis/gluconeogenesis, glycerophospholipid and purine/pyrimidine** pathway alterations. Personal rather than ambient sampling in an elderly cohort is the strength and is uncommon; against that, the ester annotations are putative, the design is cross-sectional with a single 72-h exposure window against a single metabolome, pathway enrichment on untargeted features has a well-known false-discovery problem, and the authors state the work is hypothesis-generating. No effect sizes are reportable from the abstract.

doi: 10.3390/metabo16080553

Zinan’dala et al. 2026 Int J Mol Sci*Urinary metabolomic alterations across contrasting ambient pollution regions, Thailand*

Untargeted ¹H-NMR urinary metabolomics comparing **51 adults in Chiang Mai (high exposure)** with **52 in Songkhla (low exposure)** during a regional pollution episode, using long-term residence as the exposure proxy. PLS-DA separated the groups ($R^2 = 0.873$ but $Q^2 = 0.447$ — good fit, modest prediction, and the gap is the standard warning sign of overfitting in this method). Eight metabolites differed at $p < 0.05$, six surviving Benjamini–Hochberg: **L-tryptophan, hippuric acid, xanthine and 5-HIAA remained significant after covariate adjustment**, with L-arginine and L-cystathionine lower in the high-exposure group. The design’s fatal ambiguity is acknowledged by the authors: **region is the exposure variable**, and Chiang Mai and Songkhla differ in diet, water, altitude and season as much as in PM_{2.5}. Hippuric acid in particular is a dietary polyphenol marker before it is anything else. Tier B for method, not for attribution.

doi: 10.3390/ijms27167193

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Hendrickx et al.	Effects of recent air pollution exposure on children's neurocognitive functions and the possible link with endothelial function	<i>Int J Environ Res Public Health</i>	Observational - cohort	Indoor and outdoor PM over 4 h, 24 h and 1 week windows; monthly NO ₂	Belgium
Wu et al.	Early-life exposure to particulate matter and sleep in Chinese toddlers: the roles of particle size and particulate constituents	<i>Toxics</i>	Observational - cohort	Early-life PM ₁ , PM _{2.5} , PM ₁₀ and PM _{2.5} constituents from high-resolution grids	China
Cardiovascular & metabolic					
Correa Lara et al.	Ambient particulate matter exposure and risk of venous thromboembolism: an updated systematic review and meta-analysis	<i>Thromb Haemost</i>	Review / synthesis	Long- and short-term PM _{2.5} and PM ₁₀ , harmonised to 10 µg/m ³	Global / multi-region
Ramirez-Moreno et al.	Regional disparities in stroke hospitalization and mortality associated with long-term fine particulate matter exposure: a nationwide ecological study in Spain	<i>Med Sci (Basel)</i>	Ecological	Regional annual mean PM _{2.5} (MITECO), region-year resolution	Spain
Yang et al.	Endothelial function in relation to low-level chronic residential air pollution in a general population: a cohort study	<i>Blood Press</i>	Observational - cohort	6-year residential PM ₁₀ 15.9, PM _{2.5} 12.8, NO ₂ 14.3, BC 1.04 µg/m ³ (spatiotemporal interpolation)	Belgium
Reproductive & developmental					
Chen et al.	Prenatal exposure to size-fractionated particulate matter and fetal growth trajectories: mediation by maternal glucose metabolism in a birth cohort	<i>Toxics</i>	Observational - cohort	Residential PM ₁ , PM _{2.5} , PM ₁₀ from 1 km spatiotemporal models	China
Respiratory & allergic					
Morvan et al.	When air pollution meets comorbidities: evidence on the causal effect of PM _{2.5} on COVID-19 hospitalizations and deaths	<i>Int J Epidemiol</i>	Spatial econometric model	Municipal weekly PM _{2.5} , instrumented by wind direction (2SLS)	France
de Souza et al.	Wildfire smoke, PM _{2.5} exposure and respiratory morbidity in a densely populated urban setting: spatial mapping and time-series evidence	<i>Sci Total Environ</i>	Time-series	Weekly mean CETESB PM _{2.5} ; INPE active-fire counts	Brazil
Mechanistic toxicology					
Moonwiriyaikit et al.	Establishment of a STAT6 reporter assay for screening environmental toxicants affecting allergic airway inflammation	<i>J Vis Exp</i>	Toxicological (in vitro)	PM and B[b]F as test constituents; no ambient exposure metric	Not stated
Nikaido-Landry et al.	Epidermal growth factor receptor dysregulation and climate change: insights into pollution-promoted lung cancer	<i>Philos Trans R Soc Lond B Biol Sci</i>	Review / synthesis	Ambient fine PM as a promoter in EGFR-mutant histologically normal lung tissue	Global / multi-region
Qin et al.	Engineered protein coronas reveal that apolipoprotein E enhances transport of particulate pollutants across the blood-brain barrier	<i>Environ Health (Wash)</i>	Toxicological (in vitro)	Aptamer-functionalised gold nanoparticles as PM surrogate; polystyrene and industrial-source PM	Not stated
Occupational & indoor					
Li & Zheng	Size-resolved PM ₁₀ -bound PFASs: respiratory deposition and association with oxidative stress biomarkers among industrial waste recycling workers	<i>Toxics</i>	Observational - cohort	Size-resolved PM-bound PFAS (PM _{2.1} , PM _{2.1-10} and cut-point bins); ln-ADI	China
Perna et al.	Impact of indoor air pollution on occupational exposure: rethinking the major health threat for airport workers	<i>Life (Basel)</i>	Review / synthesis	Ultrafine PM, black carbon, PAHs and trace metals in airport indoor microenvironments	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Sadki et al.	Real-time monitoring of formaldehyde exposure and acute health symptoms in healthcare workers in Morocco: a cross-sectional study	<i>Cad Saude Publica</i>	Cross-sectional survey	Shift-long air-borne formaldehyde by calibrated semiconductor dosimeter (gas, not PM)	Morocco
Sensing, forecasting & instrumentation					
Chen et al.	Study on the pollution characteristics of aerosols and VOCs along the lower reaches of the Yangtze River based on shipborne observations	<i>Atmos Environ</i>	Metadata only (no abstract)	Shipborne aerosol and VOC concentrations (not extractable)	China
Kang et al.	Emission characteristics and human health impacts of vehicle originated condensable particulate matter	<i>J Hazard Mater</i>	Multi-instrument field evaluation	Real-time condensable PM mass and number (PNU-CPM system); CPM emission factors	South Korea
Lefevre et al.	Urban microclimate dataset from low-cost sensor networks in a hot and humid tropical climate: Reunion Island	<i>Data Brief</i>	Network deployment + field evaluation	None; T, RH, wind, GHI, grey-globe temperature at 10-min cadence	Reunion Island (France)
Li et al.	Early warning of cucumber angular leaf spot by estimating airborne pathogen aerosols with particulate matter sensors	<i>Plants (Basel)</i>	Network deployment + field evaluation	Low-cost PM sensor concentration as a physical proxy for pathogen aerosol load	China
Moteki et al.	Constraining the refractive index of fractal-like aggregates from complex forward-scattering amplitudes of mass-classified aerosol particles: application to black carbon samples	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Complex refractive index of mass-classified black carbon aggregates	Not stated
Wang et al.	Perchloric acid (HClO ₄) drives atmospheric new particle formation enhanced by dimethylamine, ammonia and sulfuric acid	<i>Atmos Chem Phys</i>	Kinetic model + data reanalysis	Cluster formation rate for PA-DMA / PA-NH ₃ / PA-SA nucleation	Arctic / coastal marine
Exposure assessment & modelling					
Chen et al.	Source-specific oxidative potential of PM _{2.5} in Xi'an: roles of water-soluble metals revealed by DTT assay and interpretable machine learning	<i>Toxics</i>	Source apportionment	Volume-normalised DTT oxidative potential (DTTv) of ambient PM _{2.5}	China
Isworo & Oetari	A risk-integrated framework for industrial air quality assessment linking dispersion modeling and health-based environmental pressure indicators	<i>Environ Sci Pollut Res Int</i>	Modelling / inventory	Gaussian-plume PM _{2.5} and PM ₁₀ ; hazard-quotient-weighted AEPI	Indonesia
Li et al.	Study on the transport mechanism of surface particulate matter based on the closed earth-air model	<i>Environ Pollut</i>	Measurement campaign	PM ₁ , PM _{2.5} and PM ₁₀ in a semi-enclosed basement; earth-air flow, CO ₂ , 222Rn	China
Mainka & Mainka	Urban cycling vs. PM _{2.5} exposure: does the benefit outweigh the risk?	<i>Preprints.org</i>	Proxy validation vs personal exposure	Trip-resolved personal PM _{2.5} from low-cost sensors; inhaled dose via ventilation rate	Poland
Moryani et al.	Inhalation bioaccessibility, pollution assessment, and speciation of heavy metals in PM _{2.5} road dust from urban and semi-urban areas of Pakistan	<i>Toxics</i>	Measurement campaign	PM _{2.5} -fractionated road dust; ALF-bioaccessible Cr, As, Pb	Pakistan
Rashid et al.	Ultrafine particles and mortality: a scoping review of epidemiologic evidence and exposure assessment limitations	<i>Toxics</i>	Review / synthesis	Ambient ultra-fine particle number concentration (<100 nm)	Global / multi-region
Wu et al.	Global fire activity exacerbates extreme PM _{2.5} pollution	<i>npj Clim Atmos Sci</i>	Metadata only (no abstract)	Extreme-percentile PM _{2.5} attributable to fire activity (not extractable)	Global / multi-region
Burden, policy & mitigation					

First author	Focus	Journal	Design	Metric	Region
Acat et al.	National trends in ambient air pollutants and cause-specific mortality in Türkiye, 2018-2024: an ecological time-trend study	<i>Atmosphere</i>	Monitoring-record analysis	Unweighted national mean station PM ₁₀ (50.4 -> 41.0 µg/m ³); PM _{2.5} excluded	Türkiye
Ozmen et al.	Long-term ambient PM _{2.5} exposure and premature mortality across Türkiye	<i>Toxics</i>	Burden estimation (GBD)	Province-level annual mean PM _{2.5} , counterfactual 5 µg/m ³ (WHO AirQ+)	Türkiye
Sommar et al.	Health economic assessment of PM _{2.5} originating from residential wood combustion - a case study in northern Sweden	<i>Int J Environ Res Public Health</i>	Burden estimation (GBD)	Population-weighted RWC-attributable PM _{2.5} , 0.16 µg/m ³ (IQR 0.10-0.21)	Sweden
Zhai et al.	Global disparities in lifetime risk of lung cancer: links to socioeconomic development, smoking, and fine particulate matter from GLOBOCAN 2022	<i>Front Public Health</i>	Ecological	Country-level PM ₁ and PM _{2.5} from global environmental databases	Global / multi-region
Zhu et al.	Short-term effectiveness of urban road-cleaning operations for fugitive road dust control: evidence from TRAKER-based mobile monitoring	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	TRAKER-derived road-dust emission potential (not extractable)	Not stated
Other clinical endpoints					
Chen et al.	Untargeted metabolomics reveals metabolic perturbations in community-dwelling elderly exposed to PM _{2.5} -bound ester compounds	<i>Metabolites</i>	Observational - cross-sectional	72-h personal PM _{2.5} (UPAS V2), mean 38.06 µg/m ³ ; 20 GC-MS-prioritised esters	China
Huang et al.	Long-term exposure to PM _{2.5} -bound metals and female breast cancer incidence in New Mexico, USA	<i>Air Qual Atmos Health</i>	Ecological	Census-tract PM _{2.5} -bound Cr, Co, Cu, Mn, Hg over a 10-year pre-diagnosis window (EWPM)	United States
Zhang et al.	Plasma metabolomic signatures of air pollution and rheumatoid arthritis risk	<i>Front Public Health</i>	Observational - cohort	Residential annual PM _{2.5} , PM ₁₀ , NO ₂ , NO _x and a composite air pollution score	United Kingdom
Zinan'dala et al.	Urinary metabolomic alterations associated with contrasting ambient air pollution exposure in Thai adults: an untargeted 1H-NMR study	<i>Int J Mol Sci</i>	Observational - cross-sectional	Regional residence as proxy for PM _{2.5} contrast (Chiang Mai vs Songkhla)	Thailand

Provenance and method

Entry window **2026-08-27 to 2026-08-27**, a single day of deposits, built on the 28 August run under the 22:00 rule. **Sources.** PubMed E-utilities on [Date - Entry], run as two separate queries (health axis 19, instrumentation axis 5, 2 overlapping). Europe PMC CREATION_DATE 31, including one Preprints.org record. Crossref by-ISSN across the tracked non-PubMed-indexed titles, 43 deposits. **OpenAlex returned HTTP 429 on top of its existing 403 date-filter failure** and contributed nothing; arXiv contributed nothing new. A **Consensus** sweep on the forward-scattering / refractive-index axis returned only the 2023 antecedent work and no in-window record, so no backfill section appears in this issue. **Screening.** 98 raw records, 95 unique after PMID/DOI/title collapse, 6 already in seen.json, **36 retained and 53 rejected** with reason logged. Rejections were dominated by a single-day *Meas Sci Technol* volume release (13 records, none about aerosol — bearing diagnostics, optical flow, ultrasonic weld simulation), four Elsevier retraction notices, and non-airborne environmental chemistry (soil, sediment, water, wildlife mercury). Two records were rejected on scope after reading: *Gaseous air pollutant exposure and atherosclerosis* (IJMS), which carries no particulate exposure metric, and *Air pollution-induced oxidative stress after CABG* (Antioxidants), a conceptual perspective with no exposure measurement. **Four abstract-free deposits** are carried at tier C on title, journal and method class alone, and are labelled as such in the digest. **Effect estimates.** Thirteen from five studies. Excluded by design: the Guangzhou toddler-sleep betas (negative hour differences, not drawable on a log-ratio axis), the Türkiye OLS slope (an absolute rate difference), the Moroccan formaldehyde AORs (a gas-phase contrast) and the NO₂/NO_x/APS metabolic-signature hazard ratios (not particulate contrasts). **Harvester change made this run.** Morocco, Pakistan, Indonesia, Reunion, Belgium and the Arctic were added to the geography substrings map in plots.py; all six were falling through the unmapped diagnostic into *Global / multi-region*. **Labels.** Subtopic, design, particle-metric and geography are single-label by dominant contribution — a judgement, not a controlled vocabulary. All DOI links were resolved before build.